TG

UNDERGRADUATE PROJECT PROPOSAL

TankGuard

An IoT-Enabled Solar-Powered Water Quality Monitoring System for Household Water Storage Tanks

ESP32 | IoT Sensors | Solar Power | Local Web Dashboard

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1. Abstract

Rooftop and ground-level water storage tanks are the most neglected link in the household drinking-water chain. Although the water entering them from the municipal supply is treated and disinfected, the water that *sits* in a tank degrades over time: the chlorine residual dissipates, sediment settles on the floor, biofilm grows on the walls, and microbial populations rise—so the water coming out of the tap is frequently very different from the water that came in. Because tanks are often forgotten, this degradation is discovered only after it affects health or causes foul odour and taste.

This proposal presents **TankGuard**, an IoT-enabled, solar-powered system that continuously measures the key water-quality parameters of water stored in a household tank—pH, turbidity, total dissolved solids (TDS) and temperature, together with water level—and computes a single *TankScore* that answers one question conclusively: **“is it time to clean this tank?”** The device is wall/lid mounted, powered by a solar panel and Li-ion battery, stores ninety days of history locally, and exposes a phone-friendly web dashboard over the household WiFi network with no cloud service required. The prototype targets a single household tank from low-cost, off-the-shelf analogue sensor modules within a tight budget, with a design that scales to multiple tanks and a custom PCB in a later stage.

Keywords: IoT, water quality, ESP32, open-source hardware, smart home, turbidity sensor, solar power, sustainable development.

2. Introduction and Motivation

Safe drinking water is a fundamental requirement for public health, but its safety is not guaranteed solely at the treatment plant. In urban and semi-urban homes, water is typically stored in overhead or underground tanks that are topped up irregularly from municipal lines or borewells. WHO identifies storage as a critical recontamination point: "the safe water chain breaks at the point of storage" unless tanks are kept clean and covered [1][2]. Practically, household tanks are opened rarely—cleanings happen on the order of months to years—and by the time the water smells or tastes bad, the tank has already become a growth reservoir.

The motivation for this project is straightforward: **tank cleaning is forgotten because there is no cheap, automatic way to know when cleaning is due**. Current practice is calendar-based guesswork ("clean every N months") or reactive (clean only after the water becomes unusable). A low-cost sensor node that monitors the actual chemical and biological state of the stored water converts guesswork into evidence, and can alert the owner precisely when the tank needs attention—so that cleaning effort is spent exactly when it is needed.

TankGuard is designed around three principles:

- **In-tank, not in-pipe:** the sensors sit in the stored water, because the problem to measure is what happens while water is stored.
- **Local and private:** all data, history and alerts live on the device and are served on the home WiFi—no cloud account, no subscription, no data leaving the home.
- **Self-powered and low-maintenance:** solar panel + battery, deep-sleep power management, cost-effective calibrated analogue probes, and a cleaning recommendation engine.

3. Problem Statement

Household water storage tanks are rarely cleaned in time because there is no practical, affordable and automatic indicator of when cleaning is required. As a result:

- Stored water loses disinfectant residual and becomes biologically re-contaminated during storage.
- Sediment and sludge accumulate on the tank floor; biofilm forms on submerged surfaces.
- Consumers detect the problem only through odour, taste, or visible discolouration—after contamination has already occurred.
- Manual cleaning is irregular, inconsistent, and wastes large volumes of water because it cannot be scheduled against actual tank condition.

Cleaning is a manual, labour-intensive chore with no widely available, safe way to carry it out automatically *once* the need is detected. This project therefore focuses on the first, decisive problem: bringing evidence to the decision. **(P1)** continuous, evidence-based detection of "tank needs cleaning".

4. Objectives

1. Design a solar-powered, wall/lid-mounted IoT node based on an ESP32 that measures pH, turbidity, TDS, temperature and water level of stored tank water.

2. Develop a reproducible calibration procedure for each sensor so that readings remain accurate over months of continuous submersion.
3. Implement a deep-sleep based power-management scheme that allows the node to run indefinitely from a solar panel and a Li-ion battery.
4. Design the *TankScore* decision engine—a weighted, trend-aware indicator that classifies tank state as GOOD, WATCH or CLEAN-REQUIRED, and tracks days since the last confirmed cleaning.
5. Build a local, phone-accessible web dashboard (no cloud) showing live readings, history graphs, status and a "Tank cleaned" confirmation control.
6. Validate the complete system on a real household tank through a multi-week soak test, and document calibration drift and maintenance.

5. Literature Review

IoT water-quality monitoring is an active research area. Sharma et al. describe an ESP32-based system integrating pH, turbidity, TDS and temperature sensors with on-device machine learning and cloud integration, reporting stable readings over 24-hour operation and low repeatability error for pH; notably, the sensing array is assembled from discrete analog probes [3]. The work reported in the *International Journal of Computer Science and Publication* demonstrates an ESP32 + Adafruit.IO design measuring the same four parameters, and provides a useful calibration methodology with buffer solutions (pH 4.0, 7.0, 10.0) and standard TDS/turbidity samples [4]. Chavhan et al. extend this to industrial wastewater with a multi-parameter suite including dissolved oxygen, electrical conductivity and water level, validated against laboratory instruments [5]. The *HydroSense* framework proposes a dual-microcontroller topology—an Arduino dedicated to precision analog capture (pH, DO) and an ESP32 handling connectivity—with dissolved oxygen probes of the galvanic type that require no polarisation time [6].

First, most systems transmit to public clouds and offer no local storage or offline dashboard. Second, the analogue probe modules that keep a prototype cheap drift and benefit from regular recalibration; production hardening would migrate to industrial RS485/Modbus probes with automatic temperature compensation and IP68 enclosures [7]. Third, none of the surveyed systems addresses the *cleaning* decision itself—they report quality values but not an actionable "clean now" trigger. This proposal directly targets those gaps: an in-tank ESP32 node built from low-cost analogue sensor modules, with a local web dashboard and a decision engine that turns continuous readings into a single, conclusive cleaning verdict.

Several open-source projects already exist online and validate the ESP32 + pH/TDS/turbidity approach this proposal adopts: *water-quality-monitor* (ESP32 with pH, EC, TDS, turbidity and temperature feeding a Blynk dashboard) [11], an end-to-end ESP32 + Python web-dashboard system [12], a ThingSpeak-based TDS/turbidity monitor with SMS alerts built for under ₹2,000 [13], a tank-specific water-level + pH + turbidity monitor with pump control [14], and a cloud (AWS)-based monitor with ML-gated alerts [15]. None of them couples solar autonomy with a fully local dashboard and a cleaning-decision engine, which is exactly the gap TankGuard fills.

6. Scope and Justification

The project scope is a **single prototype system for one household tank**, built first from off-the-shelf modules on a breadboard to validate sensing and firmware, with a later custom PCB and IP65 enclosure. The design deliberately chooses:

- **Low-cost analogue sensor modules** (Gravity/Dual-head-class) in place of industrial RS485/Modbus probes: they are a fraction of the price, breadboard-friendly, and accurate enough for a validated

prototype when calibrated against handheld meters. Drift is handled by a quarterly recalibration procedure, and production hardening can drop in industrial probes later.

- **Local-only connectivity.** Data never leaves the home; the ESP32 serves a web app over WiFi (mDNS hostname). This avoids recurring costs, privacy concerns and dependence on a commercial cloud.
- **Single-node architecture.** One device measures, decides, stores and serves the dashboard; no separate actuator module is within this project's scope, though the decision output is designed to drive one in a future extension.

7. System Design Overview

TankGuard is a single node installed on the tank lid/wall with probes hanging below the water line (Figure 1). It measures the stored water, evaluates whether cleaning is due, keeps a rolling local history, and surfaces everything on a phone dashboard.

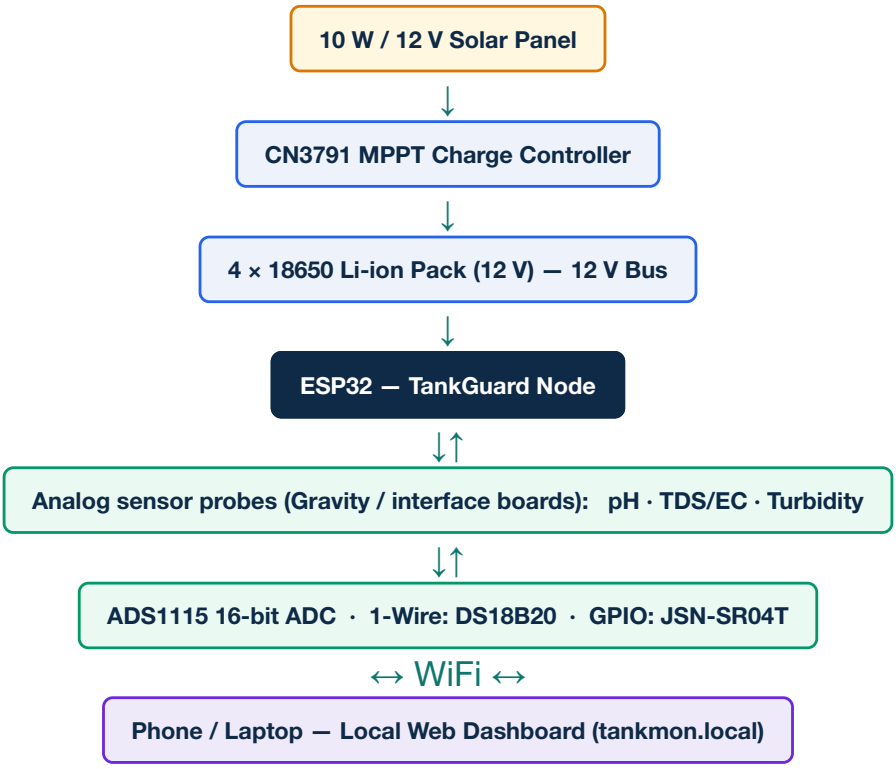


Figure 1 — TankGuard system architecture (solar battery node + local dashboard).

7.1 Modular System Product Line

TankGuard is architected as a **two-module product family around a single base unit**, so each household buys only the features it actually needs: the base unit (M1) monitors water quality, and one optional plug-in module (M2) adds safe pump control. The customer selects the combination at purchase.

Module	What it does	When a customer chooses it
M1 — Water-Quality Monitoring Module (base unit)	Continuous in-tank sensing of pH, turbidity, TDS, temperature and water level; computes the TankScore; serves the local dashboard, history and cleaning alerts.	**Every system** — this is the core around which the others plug in, and the scope of this prototype.
M2 — Auto-Off / Auto-On Pump Module	Opto-relay + motor-rated contactor with local TFT and one-button override; cuts the pump when the tank is full (level trigger) or water quality is unsafe, restarts on demand, and protects the motor from dry-running.	Wanted: pump automation driven by the same monitoring verdict.

Module options for the TankGuard product line (customer-selectable).

Module M2 is **outside the tank-side prototype's build scope** (the prototype covers the M1 base only, preserving the single-node guarantee in Section 6). The base still provides the *decision signals* for it today: the TankScore, level and alert triggers are exposed on an expansion header and emitted as dashboard/POST events, so M2 can be added later by plugging in the home board and enabling the module in the configuration file—no sensor changes required.

7.2 Monitoring Node

The node is a wall/lid-mounted **waterproof, heat-protected enclosure** (IP66 box with a foil-foam thermal liner and a reflective heat shield for exposed rooftop mounting in Indian sun) containing the ESP32, the ADS1115 ADC, power electronics, and a humidity-sealed cable gland through which the probe cables pass to the water. A full table of sensor choices and their rationale is given in Table 1. All three primary liquid sensors are low-cost analogue modules: pH, TDS/EC and turbidity are read as small analogue voltages by their interface boards and digitised by a dedicated ADS1115 16-bit ADC, which keeps them well clear of the ESP32's native ADC nonlinearity. This is the key prototype trade-off—analogue modules cost a fraction of industrial RS485 probes, and accuracy is secured by a reproducible calibration procedure validated against a handheld meter during bring-up.

Parameter	Sensor / Interface	Safe range (WHO)	Why it matters in-tank
Turbidity	Analog optical (SEN0189/Gravity), via ADC	< 1 NTU	**Primary sludge indicator;** rises with sediment and biofilm release.
pH	Analog glass electrode (SEN0161/ Gravity), via ADC	6.5–8.5	Drift signals algae, decay, seepage or corrosion.
TDS / EC	Analog conductivity (SEN0244/ Gravity), via ADC	< 500 ppm	Dissolved solids leach from settled debris and scale.
Temperature	DS18B20 waterproof, 1-Wire	—	Biological activity clock; temp-compensates other readings.
Water level	JSN-SR04T ultrasonic, waterproof	—	Water age since refill; empty/full detection.
Battery / solar	ADC voltage divider	—	State-of-charge and solar health for self-diagnostics.

Table 1 — Parameter selection and rationale.

7.3 The "When to Clean" Quality Engine (TankScore)

The decision engine converts raw readings into one actionable status. It combines two signals:

1. **Interval counter:** days since the last confirmed cleaning (set via the dashboard) versus a configurable interval (default 90 days, adjustable 30–180). Expiry alone raises a WATCH flag.
2. **Water-quality score (0–100):** each parameter contributes a sub-score based on how far the reading is from its safe band (Table 2), weighted by importance. A 14-day slope of turbidity, TDS and pH detects *deterioration in progress* even before a hard threshold is breached (e.g. turbidity climbing steadily but still under 1 NTU).

Parameter	GOOD	WATCH	CLEAN-REQUIRED flag
Turbidity (NTU)	< 1	1–5	> 5
TDS (ppm)	< 300	300–500	> 500
pH	6.5–8.5	6.0–6.5 / 8.5–9.0	< 6.0 or > 9.0
Water temperature	Not flagged directly; warm water (> 30 °C) raises the risk weight of the biological indicators.		

Table 2 — Quality thresholds driving the TankScore.

The weighted result is a single status: **GOOD** (no action), **WATCH** (deteriorating or near thresholds), or **CLEAN-REQUIRED** (recommends that the tank be cleaned). The status is displayed prominently on the dashboard. When CLEAN-REQUIRED is reached, a persistent alert is raised; the dashboard's one-tap "Tank cleaned" control logs the completed cleaning and restarts the interval counter, so the next "due" alert begins a fresh period.

7.4 Local Data and Web Dashboard

All history is stored on the ESP32 in its flash filesystem (approximately 90 days at a 3-hour sampling interval in a ring buffer). The ESP32 serves a single-page web application—gauge cards per parameter, history charts, battery/solar status, the TankScore banner, a "Tank cleaned ✓" button, and calibration screens. The phone simply opens `http://tankmon.local` when on the same WiFi network. On first boot, the device opens a captive-portal access point so the owner can enter WiFi credentials once. No cloud service, account, or subscription is required.

7.5 Power and Energy Management

The 12 V solar bus powers the sensor probes; a buck converter and a low-quiescent LDO feed a clean 3.3 V rail for the ESP32. To keep average consumption low, the node spends most of its time in deep sleep (timer wake), waking every 3 hours to power the sensor rail, let probes settle, sample, store, and publish over WiFi, then sleeping again. The RTC and a battery-monitor ADC stay alive. This measure–sleep–publish loop is the enabler for a fully solar-autonomous node (Section 11 gives the numbers).

8. Hardware Architecture

The detailed wiring follows a small number of buses (Table 3). A printed schematic and a KiCad/Fritzing prototype layout will accompany the firmware repository; the breadboard bring-up plan verifies each sensor in isolation before integration.

Interface	ESP32 pins	Connected device
I2C	GPIO21 / GPIO22	ADS1115 16-bit ADC — pH, TDS and turbidity analogue channels
1-Wire	GPIO4	DS18B20 waterproof probe
Ultrasonic trigger / echo	GPIO25 / GPIO26	JSN-SR04T water-level sensor
ADC (attenuated)	GPIO34 (channel)	Battery voltage divider
Power switch (sensor rail)	GPIO27	High-side MOSFET on 5 V analogue sensing rail
RTC wake source	—	Timer wake from deep sleep (3 h default)
Expansion header (M2)	GPIO32 / GPIO33 / GPIO14	Trigger outputs for the optional Pump-Control (M2) module

Table 3 — Preliminary pin and bus plan (ESP32-WROOM-32).

> **Prototype build strategy:** Stage A — breadboard the ESP32 + ADS1115 + independent drivers for each analogue sensor and validate range/calibration against a handheld meter; Stage B — integrate all probes and interfaces inside the single waterproof, heat-protected enclosure; Stage C — add solar power train and deep-sleep; Stage D — on-tank validation, then optional custom PCB.

9. Software and Firmware Architecture

The firmware is written for the Arduino core under PlatformIO (ESP32), organised in modules so each sensor and service can be developed and tested independently:

```
main:  measure → compute TankScore → store → serve → deep sleep
├── sensors/    ph, tds, turbidity, temp, level, battery drivers
├── power/      battery ADC, sensor-rail enable, deep-sleep scheduler
├── wifi/       station + mDNS + captive-portal first-boot provisioning
├── server/     HTTP REST + WebSocket + LittleFS static web app
├── storage/    LittleFS config, calibration, 90-day history ring buffer
└── quality/    TankScore engine, trend detection, alert rules
```

Key engineering details:

- **Noise rejection:** the three analogue probes are sampled as ~100 samples each through the ADS1115, sorted, and the median used; pH and TDS are temperature-compensated in software against the DS18B20.
- **Calibration:** pH via two/three-point buffer calibration (4.0/6.86/9.18), turbidity via 0 NTU and 1000 NTU formazin standards, TDS via standard conductivity solutions; offsets are persisted in a calibration file and re-applied on every boot. A maintenance reminder re-runs calibration every 3 months.
- **Time:** NTP synchronisation whenever WiFi is online; the RTC keeps time in deep sleep.
- **Analogue channels:** pH, TDS and turbidity are sampled through the ADS1115 to bypass the ESP32's native ADC nonlinearity; like every channel, they are median-filtered and temperature-compensated, and gated by the sensor-rail power switch so they draw current only during the measurement window.

10. Bill of Materials (BOM)

10.1 Product-Line Bill of Materials (two modular kits)

All parts for the complete two-module product line—the M1 base unit plus the optional M2 pump-control kit—are listed in one bill (Table 4) with Indian-market prices in rupees (items marked 'est.' are typical street estimates). The full family sums to **₹14,550, deliberately under the ₹15,346 budget cap**, so no module added later breaks the plan. The prototype build purchases the M1 base kit only (items 1–20, ₹13,550); M2 is an add-on kit (items 21–23) a customer buys only when ticking that option. The enclosure is specified as **waterproof and heat-protected** for exposed rooftop mounting in Indian sun. ORP, dissolved-oxygen and free-chlorine probes (roughly ₹22,000 together at current street prices) are deliberately *deferred* and marked as future expansion, keeping the build affordable without compromising the core five-parameter verdict.

#	Category	Component	Note / SKU	Qty	Unit (₹)	Total (₹)
M1 — Water-Quality Monitoring Module (base kit; prototype scope)						
1	Compute	Microcontroller board	ESP32-WROOM-32 dev board (FireBeetle-class, DFR0478)	1	1,237	1,237
2	Compute	16-bit ADC	ADS1115 module (4-analogue-channel expansion)	1	145	145
3	Sensing	pH sensor kit	Gravity analogue pH (SEN0161-V2), probe + interface board	1	2,669	2,669
4	Sensing	TDS / EC sensor	Gravity analogue TDS (SEN0244)	1	1,489	1,489
5	Sensing	Turbidity sensor	Gravity analogue turbidity (SEN0189)	1	600	600
6	Sensing	Temperature probe	Waterproof DS18B20 (stainless, 1 m cable)	1	44	44
7	Sensing	Water-level sensor	JSN-SR04T waterproof ultrasonic	1	270	270
8	Power	Buck converter	12 V → 5 V DC-DC (LM2596S)	1	78	78
9	Power	3.3 V LDO	Low-quiescent HT7333 (ESP32 rail)	1	65 est.	65
10	Power	Sensor-rail switch	MOSFET high-side enable boards (IRF520)	2	42	84
11	Power	MPPT charge controller	CN3791 solar module	1	420	420
12	Power	Solar panel	10 W / 12 V polycrystalline	1	950 est.	950
13	Power	Battery cells	18650 Li-ion (~3000 mAh each)	4	190 est.	760
14	Power	Battery protection	3S BMS with balance	1	420 est.	420
15	Enclosure	Waterproof housing		1	900 est.	900

			IP66 weatherproof box, cable glands, mounting plate, sealed cable entry			
16	Enclosure	Heat-protection set	Foil-foam thermal liner, reflective heat shield, desiccant, vented baffle	1	620 est.	620
17	Build	Wiring & misc	Connectors, divider resistors, fasteners, potting compound	1	739 est.	739
18	Calibration	pH buffers	4.00 / 6.86 / 9.18 buffer solutions (set)	1	580 est.	580
19	Calibration	Turbidity standards	0 NTU (DI water) + 1000 NTU formazin (set)	1	480 est.	480
20	Spares	Probe care & spares kit	KCl storage solution, spare caps and seals, backup electrode	1	1,000 est.	1,000
M1 subtotal (base kit, prototype)						**13,550**
M2 — Auto-Off / Auto-On Pump Module (optional add-on kit)						
21	M2	Pump relay kit	Opto-isolated 5 V relay + 9 A AC-3 contactor	1	550 est.	550
22	M2	Status display	1.8″ ST7735 TFT + one override button	1	250 est.	250
23	M2	Mains parts	Terminal block, rated flex wire, MCB, dry-run guard	1	200 est.	200
M2 subtotal (optional kit)						**1,000**
Total parts cost — full two-module product line						**14,550**

Table 4 — Product-line BOM by module (verified Indian-market pricing, parts only; ‘est.’ = street estimate).

Module	Amount (₹)
M1 — Water-Quality Monitoring Module (base kit, prototype)	13,550
M2 — Auto-Off / Auto-On Pump Module (optional kit, items 21–23)	1,000
Total parts cost (full two-module product line)	**14,550**

Table 5 — Budget summary by module.

The total is **parts-only**—the sum of the 23 items above, excluding labour, GST, freight and import duties. At ₹14,550 it sits **under the ₹15,346 budget cap**, and the prototype’s build scope (M1 base kit) costs just ₹13,550. Prices marked ‘est.’ are typical Indian street estimates and should be confirmed at purchase; the remainder are verified prices compiled from Indian electronics retailers during the market survey. A nominal contingency of ~10% (≈ ₹1,455) would bring the delivered full-family unit to about ₹16,005. ORP, dissolved-oxygen and free-chlorine probes remain a future expansion and are excluded from this budget.

11. Power Budget and Solar Sizing

Energy autonomy is the most important reliability requirement. Table 8 summarises the projected consumption of the measure–sleep–publish loop.

State	Current (avg)	Duty / day	Charge used / day
Deep sleep (ESP32 + RTC only, rail off)	~0.1 mA	22 h	2.2 mAh
Measure window (analogue probes settle, ADS1115 sampling)	~90 mA	1.33 h (4 min × 8 wake-ups)	120 mAh
WiFi publish + NTP	~250 mA	1.5 h (approx. 12 min/ wake-up)	375 mAh
Total average daily consumption			**~500 mAh @ 12 V**

Table 8 — Power budget (Monitoring Node).

With a 4×18650 pack (~10.8 Ah usable at 12 V), the node has roughly **3 weeks of autonomy with zero sun**—far exceeding the worst realistic cloud cover. A 10 W panel yields ~50 Wh/day at 5 peak-sun-hours, comfortably recharging a 6 Wh/day draw even in winter. The analogue probe boards (pH, TDS, turbidity) draw only a few mA each and are switched with the rest of the sensor rail, so the daily budget is essentially unchanged.

12. Expected Outcomes

- A working, solar-powered prototype Monitoring Node producing calibrated readings for pH, turbidity, TDS, temperature and water level, with ≤3-minute settle time per measurement window.
- A base module exposing TankScore and level triggers on an expansion header so the optional pump-control (M2) module bolts on without sensor changes.
- A validated TankScore engine that correctly identifies GOOD / WATCH / CLEAN-REQUIRED states on a real tank, including trend-based early warnings.
- A local web dashboard (no cloud) with live gauges, 90-day history graphs, battery/solar telemetry, and one-tap "Tank cleaned" confirmation.
- Documented calibration procedure, probe-drift measurements from a multi-week soak test, and a maintenance guide (membrane/KCl care, quarterly recalibration).

13. Work Plan and Timeline

Weeks	Activity	Milestone / Deliverable
1–2	Literature review; vendor survey; finalise BOM; order parts	Approved BOM; parts received
3–4	Breadboard bring-up: validate each sensor standalone; calibrate with standards	Per-sensor calibration records
5–6	Firmware skeleton: deep-sleep scheduler, battery monitor, solar charge train	Sustained solar-only operation
7–8	All sensor drivers; median filtering; temperature compensation; TankScore engine	Complete reading + scoring
9	Captive-portal WiFi provisioning; HTTP/WebSocket server; LittleFS history; dashboard	Remote dashboard live at tankmon.local
10–11	On-tank installation and commissioning; multi-week soak test with drift logging	Validated field performance
12	Drift analysis, validation report, documentation, demo and defence	Final report and presentation

Table 9 — 12-week indicative schedule.

14. Risks and Mitigations

Risk	Likelihood	Impact	Mitigation
Probe fouling / calibration drift over weeks	High	Medium	Waterproof probe heads, quarterly recalibration, probe-care (KCl storage), drift logging.
Solar variability (winter / cloudy periods)	Medium	Medium	Over-sized battery (~3 weeks autonomy), deep-sleep economy, optional panel upgrade path.
ESP32 ADC non-linearity on analogue channels	Medium	Low	Dedicated ADS1115 16-bit ADC for all analogue probe channels, calibrated against a handheld meter.
WiFi not reachable at tank location	Medium	Low	Captive-portal fallback access point; data is never lost (local flash ring buffer).
Enclosure condensation / moisture and peak-sun heat	Medium	Medium	IP66 box + glands, heat-reflective housing and foam insulation, desiccant, sealed cable entry.

Table 10 — Key engineering risks.

15. Conclusion

TankGuard addresses a real, neglected problem—the silent degradation of household stored water—with a practical, evidence-based, fully local and solar-autonomous system. By measuring the parameters that actually change while water sits in a tank and computing a single TankScore, plus tracking how long it has been since a confirmed cleaning, the device moves tank maintenance from guesswork to a clear, timely schedule. The project follows a measured path—validated sensor bring-up, robust firmware, local

dashboard, then real-tank soak test—producing a documented, reproducible prototype whose design can later be hardened into a custom PCB and sealed product.

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